MILITARY SPECIFICATION

SEMITRAILER, STAKE: COMMERCIAL, DROP FRAME, TANDEM AXLE, 20 TON

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of commercial tandem axle drop frame type stake semitrailer of 20 ton minimum payload capacity.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, and standards. The following specifications, and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the soliciation.

SPECIFICATIONS

FEDERAL

ZZ-T-381 - Tires, Pneumatic, Vehicular (Highway).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 2330

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MILITARY

MIL-W-3912 - Wood Parts, Fabricated: for Transport Vehicle Bodies and Similar Applications.

STANDARDS

FEDERAL

FED-STD-297 - Rustproofing, of Commercial (nontacticle) Vehicles.
FED-STD-595 - Colors.

MILITARY

MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281 - Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-1223 - Nontactical Wheeled Vehicles Treatment, Painting, Identification Marking and Data Plates Standards.
MS75021 - Connector, Receptacle, Electrical - 12 Contact, Intervehicular, 28 Volt, Waterproof.

2.1.2 Other Government documents, and publications. The following other Government documents, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DEPARTMENT OF TRANSPORTATION (DoT)

Federal Motor Carrier Safety Regulations.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION, INC. (TRA)

TRA Yearbook.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First production vehicle. When specified (see 6.2), the contractor shall furnish one complete semitrailer for first production vehicle inspection (see 4.2.1 and 6.4).

3.2 Standard commercial product. The equipment shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the equipment being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model(s).

3.3 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and

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reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.4 Interchangeability. All units of the same classification furnished with similar options under a specific contract shall be identical to the extent necessary to insure interchangeability of component parts, assemblies, accessories, and spare parts.

3.5 Special requirements. In addition to standard semitrailer and components specified in 3.2, semitrailer shall be furnished with special equipment as specified herein.

3.5.1 Treatment, painting, identification marking, and data plates. Unless otherwise specified (see 6.2), treatment, painting, identification marking, data plates, and registration numbers shall be in accordance with MIL-STD-1223, as specified (see 6.2), by the procuring activity for the appropriate service. Color of the finish coat shall conform to FED-STD-595 as specified for the Navy. Flooring shall be painted same color as the semitrailer.

* 3.5.2 Rustproofing. Rustproofing shall be in accordance with FED-STD-297.

3.5.3 Hardwood. Flooring and other wood parts specified herein to be furnished in hardwood shall be fabricated from the types of lumber listed in table I of MIL-W-3912. Apitong is acceptable for flooring only.

3.5.4 Wood treatment. Wood surface shall be treated in accordance with the requirements of MIL-STD-1223, except manufacturer's method, which provides equivalent protection, may be substituted upon approval by the contracting officer. Apitong need not be treated.

3.5.5 Lifting and tiedown attachments. When specified (see 6.2), the semitrailer shall be equipped with lifting and tiedown attachments. Lifting and tiedown attachments shall conform to type II or type III of MIL-STD-209. A nonferrous transportation plate shall be provided and mechanically attached to the semitrailer. Transportation plates shall be inscribed with a diagram showing the lifting attachments and lifting slings, the capacity of each attachment, and the required length and size of each sling cable. A silhouette of the item furnished showing the center of gravity shall be provided on the transportation plate. Tiedown attachments may be identified by stenciling or other suitable marking. Tiedown marking shall clearly indicate that the attachments are intended for the tie down of the semitrailer on the carrier when shipped.

3.5.6 Center of gravity. When specified (see 6.2), the longitudinal center of gravity of the unloaded semitrailer shall be located and permanently indicated in a manner to be defined by the procuring activity.

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3.6 Weights and loads.

3.6.1 Net weight. The net weight of the fully equipped semitrailer shall be the minimum practicable for the service intended.

3.6.2 Rated payload capacity. The rated payload capacity, evenly distributed over the load space, shall be not less than 40,000 pounds (lb). The load space is defined as the total length of the platform including both lower deck and kickup (gooseneck) deck.

3.6.3 Gross weight. The gross vehicle weight (gvw) shall consist of the net weight and the rated payload evenly distributed over the load space.

3.6.4 Weight distribution. Load bearing members shall be rated for the loads imposed with the frame in a horizontal position, and the payload evenly distributed over the load space. The proportion of the gvw to be supported on the axle suspensions, measured at each wheel at the ground shall be approximately equal to, but not more than, 32,000 lbs.

3.7 Performance. The semitrailer, fully equipped and loaded with specified payload (see 3.6.2), shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unimproved roads (see 6.6.2), and over reasonably hard uneven terrain; and being towed at speeds as great as 55 mph over improved roads (see 6.6.1).

3.7.1 Turning ability. The semitrailer shall be capable of assuming a 90 degree (°) angle to the coupled towing vehicle without cramping or damage to the semitrailer or the towing vehicle.

3.7.2 Tracking ability. The semitrailer shall conform to the tracking requirements of DoT Federal Motor Carrier Safety Regulations, section 393.70(e).

* 3.7.3 Brake Performance. The service brakes shall stop the semitrailer-tractor combination, with the semitrailer loaded with specified payload, within the stopping distance requirements of DoT Federal Motor Carrier Safety Regulation 393.52.

3.8 Dimemsions and clearances. The semitrailer shall be a two-level type having a lower main deck and a full width kickup (gooseneck) deck. The semitrailer shall conform to the dimensions specified in table I while resting on the landing legs on level ground, uncoupled from the towing vehicle, without payload.

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Table I. Dimensions and clearances.

Overall length, feet ±1 inch	40
Lower deck length, feet (not less than)	29
Overall width, inches (-1 inch)	96
Ground clearance, inches (not less than)	9-1/2
Landing gear clearance	see 3.9.8
Turning clearance (from the centerline of kingpin to any portion of semitrailer 6 inches or more below upper fifth wheel plate), inches (not less than)	82
Swing radius from centerline of kingpin to most distant point on semitrailer nose, inches (not more than)	58
Distance from centerline of kingpin to front end of semitrailer, inches (not less than)	30
Upper fifth wheel height (from the ground to underside of plate), inches ±1 inch	47
Lower deck height, inches (not more than)	45
Drop height, inches (not less than)	12

* 3.8.1 Stress analysis. Stress analysis for the complete semitrailer, (including gooseneck), loaded as specified in 3.6.2, shall be furnished. The stress analysis shall include shear and moment diagrams and deflection calculations. The strength of the floor, cross members, and outside frame members shall not be included in the calculations to compute the main frame maximum fiber stress. But the weight of the floor, cross members, and outside frame members imposed on the main frame shall be included in the total load imposed on the main frame.

3.9 Chassis components. The chassis components shall be the manufacturer's current standard construction, adequate to support the maximum specified payload (see 3.6.2), under the operating conditions specified herein.

3.9.1 Frame construction. The frame shall be of the drop type, fabricated of high tensile steel having not less than 50,000 pound-force per square inch (psi) tensile strength. The main frame rails shall be of high tensile steel I-beam construction. The kickup frame and lower deck frame shall be reinforced and welded into a single structure. The crossmembers shall be not more than 2 feet apart, except over wheels. The specified payload (see 3.6.2), and the sprung weight of the semitrailer shall not impose a fiber stress greater than 50 percent of the yield strength of the material used, under all conditions of loading up to rated payload capacity. The kickup wall shall be perpendicular to the deck plane.

3.9.2 Suspension system. The semitrailer shall be furnished with the manufacturer's standard suspension system. Each component of the suspension system shall have a rated capacity at least equal to the load imposed, measured at the ground, when the semitrailer is loaded with its specified payload (see 3.6.2). Clearances shall preclude interference between wheels, tires, or any other part of the semitrailer, under the operating conditions specified herein.

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* 3.9.3 Axles. Axle ratings shall be at least equal to the load imposed on each axle, measured at the ground, when the semitrailer is loaded with rated payload. The wheel bearings and axle spindles shall be oil lubricated. The oil viscosity shall be in accordance with the manufacturer's recommendations. the hubcaps shall have a window for visual determination of oil level. Provisions for venting, or other method of withstanding internal pressure buildup, and for replenishing the oil supply shall be incorporated. Hubs shall be fitted with weatherproof seals.

3.9.4 Wheels, rims, tires, and tubes.

* 3.9.4.1 Wheels, rims, and tires. Semitrailer shall be equipped with dual wheels on each axle. When specified (see 6.2), disk type wheels shall be furnished. The rims and tire ratings shall conform to TRA recommendations for the type and size of tires furnished. Tire and rim sizes shall be the same for all wheels on the semitrailer. Tires shall be tube- or tubeless-type with highway tread and shall meet outside diameter dimensional requirements of ZZ-T-381. Tires shall be of rated capacity at least equal to the load imposed on each tire, measured at each wheel at the ground, when the semitrailer is loaded with rated payload. Manufacturer's current size and type tires, conforming to requirements specified herein, may be furnished.

3.9.4.2 Tire carrier. One tire carrier shall be provided and installed under the frame at cub side in an accessible location. Means shall be provided for securing the tire within the carrier to prevent accidental loss.

3.9.4.3 Inner tubes. When tube-type tires are furnished, inner tubes shall be heavy-duty type, and shall be of proper size for the tires furnished. Tire flaps shall be provided in accordance with TRA recommendations.

3.9.4.4 Spare wheel. Unless otherwise specified (see 6.2), a spare wheel or rim shall be mounted on the tire carrier.

3.9.4.5 Spare tire. When specified (see 6.2), an inflated spare tire shall be mounted on the spare wheel or rim specified in 3.9.4.4. The spare tire shall be of the same size and tread design and of the same ply rating as tires installed on the semitrailer.

* 3.9.5 Brakes. Brakes shall be of the full air, and shall conform to DoT Federal Motor Carrier Safety Regulations, sections 393.40 through 393.43, 393.45 through 393.47, and as specified herein. The braking system shall include standard breakaway features, air reservoir, manual or automatic slack adjusters, piping, hose connections, gladhands, spring loaded dust covers or dummy gladhands equipped with security chains, and all other components required for a complete air-brake system. Gladhands shall conform to SAE J318. The braking system shall be installed in a manner which provides road clearance for travel over uneven terrain and protection against damage caused by objects striking components. No part of the braking system shall extend below the bottom of wheel rims.

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* 3.9.5.1 Parking brakes. Spring or air diaphragm mechanical lock-type parking brakes shall be furnished. The parking brakes shall be automatically applied upon disconnection of the emergency air line and under emergency braking conditions. The parking brakes shall remain in the applied condition with no additional application and despite the depletion of all air pressure. Parking brakes shall conform to DoT Federal Motor Carrier Safety Regulation 393.41.

3.9.6 Upper fifth wheel plate. The upper fifth wheel plate shall be designed for coupling to a full oscillating and fore and aft rocking type fifth wheel, and shall be of sufficient size to cover a fifth wheel 36 inches in diameter, and shall conform to DoT Federal Motor Carrier Safety Regulations, section 393.70(b). The kingpin shall be fabricated of heat treated alloy steel and shall conform to SAE J700. The forward end of the upper fifth wheel plate or skid plate shall have a turned up lip for ease of loading and for body protection.

* 3.9.7 Lighting. Unless otherwise specified (see 6.2), the electrical lighting system shall be 12-volt (V) potential. The lighting system shall comform to DoT Federal Motor Carrier Safety Regulations, sections 393.14, 393.20, 393.22, 393.23, 393.25 through 393.29, 393.32, and 393.33. All lights and reflectors shall be protected from operational hazards by mounting in recessed or otherwise guarded locations. Lights and reflectors shall not be mounted on vertical surfaces of the rub rails or semitrailer bumpers. The front of the semitrailer shall be equipped with a receptable conforming to SAE J560, with the conductors connected and color coded as specified therein. The receptacle shall be located in accordance with SAE J702. All electrical wiring shall conform to SAE J555.

3.9.7.1 Receptacle system 24V. When specified (see 6.2), the semitrailer shall be equipped with one 12-contact receptacle with cover and one 7-contact receptacle with cover installed at the front of the semitrailer. The 12-contact receptacle shall conform to MS75021, part No. MS75021-1. The 7-contact receptacle shall confrom to SAE J560 round socket for jumper cable. The 12-contact receptacle shall be provided with resistance to each circuit to reduce the voltage of a tactical military design towing vehicle from 24V to 12V. The 7-contact socket shall be wired in accordance with SAE J560 to supply 12V directly to the semitrailer circuits. The 24V, 12-contact receptacle shall be connected through the necessary resistors into the 12V trailer circuits as follows:

A - Do not connect.
B - Connect to circuit No. 3.
C - Do not connect.
D - Connect to circuit No. 1.
E - Connect to circuits No. 2 and 6.
F - Do not connect.
H - Do not connect.
J - Connect to circuit No. 5.
K - Connect to circuit No. 7.
L - Do not connect.
M - Do not connect.
N - Do not connect.

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Circuits B and J on tactical truck-tractors are combination stop and turn indicator circuits. The normal 12V turn signal lights will function both as turn signals and stoplights and the normal 12V stoplights will not be in operation when the semitrailer is connected to a towing vehicle with a 24V power supply. Because of this condition, the No.4 on stoplight circuit is not connected to the 24V, 12-contact receptacle.

3.9.8 Landing gear. Semitrailer shall have two vertical lift, telescoptic, nonrotating landing legs, with two speed gears and a handcrack on the curb side. Landing legs shall be equipped with steel wheels or self-leveling skid pads. When specified (see 6.2), the landing gear shall be capable of being operated from either side of the semitrailer. Supports for the crank extension shafts and clips for holding cranks, when folded, shall be provided. The landing gear shall withstand, without deformation, the combined static and dynamic forces due to proportion of gross weight sustained and the forces resulting from impact during coupling and uncoupling operations. When placed in travel position, the landing gear legs shall remain positively locked. The landing gear shall have a range of adjustment to vary the height of the upper fifth wheel from 47 inches to not less than 52 inches from the ground. With the semitrailer coupled to a towing tractor and in level position, the clearance under the fully retracted landing gear shall exceed the semitrailer ground clearance, but shall in no case be less than 11 inches.

3.9.9 Rear end protection. Semitrailer rear end protection shall be in accordance with DoT Federal Motor Carrier Safety Regulation, section 393.86.

3.9.10 Rear wheel splash and stone throw protection. Rear wheels shall have mud flaps at rear. Splash and stone throw protection shall be in accordance with SAE J682.

* 3.10 Lubrication. The lubricating points shall be easily visible and accessible. Hydraulic lubrication fittings shall be in accordance with SAE J534. Where use of high pressure lubricating equipment, 1,000 psi or higher, will damage grease seals or other parts, a suitable warning shall be affixed to the equipment in a conspicuous location.

* 3.11 Platform. Each deck (lower and kickup) shall be level and completely covered having no cutouts or protrusions. The platform shall have crossmembers not more than 2 feet apart, except over wheels. Protruding or nonprotruding steel stake pockets shall be furnished. Protruding type pockets, when provided, shall be protected all around by a steel rub rail not less than 2 inches wide and not less than 1/4-inch thick. When nonprotruding type pockets are furnished, the outer side rail may serve as rub rail. Front and rear shall have not less than four pockets each, and the side pockets shall be on not more than 24-inch centers. The kickup deck, which shall be the full width of the trailer, and the lower deck, shall be floored with hardwood lumber conforming to 3.5.1.2 or apitong. Laminated hardwood is not acceptable. Floorboards shall be not less than 1-5/16 inches thick, laid lengthwise with tongue-and-groove or shiplap joints. Flooring shall be secured with countersunk, corrosion resistant screws or bolts. Not less than two securing screws or bolts shall be installed at each cross sill and not less than two on each side of each butt joint. Top of countersunk screw or

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bolt head shall be not less than 1/8 inch and not more than 1/4 inch below top of wood surface. Butt joints shall be centered over crossmembers and staggered. When rub rails cannot be used for securing cargo, tiedown devices for that purpose shall be provided and spaced on centers not greater than 2 feet. Cargo tiedown devices, when required, shall be of not less than 3/8-inch diameter steel stock and shall be protected by the rub rail. The rear of the platform shall be provided with means for attaching the ramps specified in 3.11.1.

* 3.11.1 Ramps. When specified (see 6.2), detachable ramps which can be attached to the rear end of the platform shall be furnished. The ramps shall provide a 15° maximum incline and be capable of supporting, as a minimum, a fork-lift truck having the following size and weight: wheel base 60 inches, tread 30 inches, 4 inch wide tires, weight 6000 lbs plus payload 3000 lbs at 24 inch load center. Ramps shall be stowed on the semitrailer, when not in use, beneath the rear of the platform or under the platform at the sides or other convenient location.

3.12 Stake panels. When specified (see 6.2), stake panels shall be furnished. Stake panels around the main deck shall be not less than 60 inches high and the kickup panels shall be of such height that the tops of the panels shall be level with the tops of main deck panels. The front panel shall be of steel, capable of being removed when unbolted. The front panel shall be designed to sustain an end thrust of not less than 50 pounds per square foot, when interlocked with the adjacent side racks. The number of side sections shall be held to a minimum, with rack section width not exceeding 54 inches. All sections shall be easily removable, without evidence of binding or jamming. The intermediate side sections shall be interchangeable on the same side, except for the adjusting racks. Side pockets shall be not more than 24 inch centers except in suspension area. All sections shall meet and fit to the same side, except for the adjusting racks. All sections shall meet and fit so that locking devices engage and disengage freely and shall not require the use of other than small handtools. The stakes shall be either hardwood, fitted with a metal ferrule at the lower end; pressed steel not less than 0.1345-inch (U.S. revised standard gage No. 10) thick; or extruded aluminum not less than 0.094 inch thick. When the racks are installed, the slats shall be inside the load space, with the upright posts outside and all corners shall be square. The front panel and side and rear racks shall be provided with heavy-duty interlocking hardware. The rear rack sections shall be equipped with locking devices to lock racks to body. Slates used in the construction of the side and rear racks shall be at least 13/16 by 3-5/8 inch finished dimensions. The slats in the bottom half shall be spaced as close together as practicable, and the slats in the upper half shall be spread not less than 2 inches and not more than 3 inches apart. Stakes and slats, including hardware, shall be held together with bolts or other through-type fasteners at least 1/4 inch in diameter. All wood parts shall be hardwood conforming to 3.5.3.

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3.13 Servicing and adjusting. Prior to acceptance of the semitrailer by the Government, the contractor shall service and adjust the semitrailer for immediate operational use as required in the operator's manual. The servicing and adjusting shall include at least the following:

1. Inflation of all tires.
2. Adjustment of brakes.
3. Proper functioning of all lighting and electrical systems.
4. Wheel alinement.
5. Adjustment of engine to include tune-up.
6. Complete lubrication with grades of lubricants recommended for ambient temperature at the delivery point.
7. Cooling system filled to capacity with a clean solution of equal parts by volume of water and antifreeze (ethylene glycol).

The semitrailer shall be conspicuously tagged to identify the lubricants and their temperature range.

3.14 Workmanship.

3.14.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.14.2 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.14.3 Riveted connections. Rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.14.4 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the parts connected by the welds. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings.

3.14.5 Castings. All castings shall be sound and free from patching, misplaced coring, warping, or any other defect which reduces the castings ability to perform its intended function.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements specified herein and in applicable referenced documents.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First production vehicle inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First production vehicle inspection. The first production vehicle produced under the contract shall be inspected by the contractor at his plant under the direction and in the presence of Government representatives. This inspection shall include the examination of 4.3 and the test of 4.4. The purpose of the inspection shall be to determine vehicle conformity with the requirements of the contract. Acceptance of the first production vehicle shall not constitute a waiver by the Government of its right under the provisions of the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.3, the tests of 4.5, and the packaging inspection of 4.6.

4.3 Examination. Each semitrailer shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

* 4.4 First production vehicle tests. The first production vehicle shall be tested as specified in 4.4.1 through 4.4.5. The test shall be performed by the contractor under the direciton and in the presence of Government representative. Failure to pass any phase of the requred tests shall be cause for the Government to refuse acceptance of all semitrailers until corrective action has been taken.

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4.4.1 Vehicle weight. Semitrailer shall be weighed to determine net weight and distribution of net weight on fifth wheel and axles. The imposed loading on the fifth wheel and axles shall be computed using the net weight and the payload. Calculated imposed loads on fifth wheel and axles shall be utilized to ascertain that the suspension, axles, and tires furnished are of adequate capacity to meet contract requirements, and to determine conformance to 3.6.

4.4.2 Road test. Road test shall consist of coupling the semitrailer, loaded with rated payload, to a truck-tractor. The tractor combination shall be driven a distance of not less than 20 miles under the conditions specified at 3.7. At least 20 percent of the distance shall be over hard uneven terrain. At least five sudden stops shall be made from a speed not less than 20 mph. Tracking ability shall be observed to verify conformance to 3.7.2. After test, the semitrailer shall be examined for evidence of misalinement, binding, or other malfunction.

4.4.3 Turning ability test. A truck-tractor shall be coupled to the semitrailer and driven through turns up to 90° right and left to determine conformance to 3.7.1.

4.4.4 Brake tests. Semitrailer, carrying the rated payload and coupled to truck-tractor, shall be driven on smooth, dry, level pavement and brakes applied from a speed of 20 mph, to determine conformance to the stopping distance required under 3.7.3. Breakaway situation shall be simulated to verify braking ability upon breakdown.

4.4.5 Parking brake test. The semitrailer, with rated payload, shall be placed on a 10 percent grade, first headed up the grade and then headed down the grade, and tested to determine conformance to 3.9.5.1. These tests shall be conducted with the semitrailer disconnected from the towing vehicle and resting on the landing legs.

4.5 Production semitrailer tests. The contractors testing system shall, as a minimum, assure that the semitrailer conforms to the physical and dimensional requirements and is capable of meeting the performance requirements specified herein.

4.6 Packaging inspection. The vehicle shall be inspected to verify conformance to the requirements of section 5.

5. PACKAGING

5.1 Vehicle processing. The equipment shall be preserved and packaged in accordance with the contractor's standard practice. When specified (see 6.2), equipment shall be preserved, packaged, and marked in accordance with the mobile requirements of MIL-STD-281 with the level of preservation and packaging as specified (see 6.2).

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6. NOTES

6.1 Intended use. Semitrailer covered by this specification is intended for use in transporting all type of cargo. Cargo will be loaded on this semitrailer by forklifts having a 14,000 lb axle load supported by 7-inch wide tires.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When first production vehicle is required for inspection and approval (see 3.1, and 6.4).
c. When treatment, painting, marking, data plates, and registration numbers are not to be in accordance with MIL-STD-1223 (see 3.5.1).
d. Appropriate service for treatment, painting, marking, data plates and color of finish coat (see 3.5.1).
e. When lifting and tying devices are required (see 3.5.5).
f. When longitudinal center of gravity is to be located and method of indicating center of gravity (see 3.5.6).
g. When disk type wheels are required (see 3.9.4.1).
h. When spare wheel or rim is not required (see 3.9.4.4).
i. When spare tire is required (see 3.9.4.5).
j. Lighting system voltage if other than as specified (see 3.9.7).
k. When 24V receptacle system is required (see 3.9.7.1).
l. When operation of landing gear from either side of semitrailer is required (see 3.9.8).
m. When detachable ramps shall be finished (see 3.11.1).
n. When stake panels are required (see 3.12).
o. When preservation and packaging in accordance with MIL-STD-281 is required, and the level of preservation and packaging required (see 5.1).

6.3 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) and invokes the provisions of DoD Federal Acquisition Regulations (FAR) Supplement 27.410-6, the data requirements will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL (DD Form 1423) incorporated into the contract. When the provisions of DoD FAR 27.410-6 are not invoked, the data shall be delivered in accordance with the contract requirements.

6.4 First production vehicle. When a first production vehicle inspection is required, the vehicle shall be tested and should be a first production vehicle or a standard vehicle from the contractor's current inventory. The contracting officer should include specific instructions in the contract regarding arrangements for examination, test, and approval of the first production vehicle (see 3.2 and 4.2.1).

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any

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inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.6 Definitions.

6.6.1 Improved road. An improved road is a smooth, hard surfaced road, such as a concrete- or asphalt-paved highway.

6.6.2 Unimproved road. An unimproved road is an unpaved, unstabilized road with an undulating surface having occasional chuckholes and exposed rocks.

Custodian:

Navy - YD

Preparing Activity:

Navy - YD

(Project 2330-N872)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-28550C(YD)	SEMITRAILER, STRAKE: COMMERCIAL, DROP FRAME, TANDEM AXLE, 20 TON

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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